

CS212 Assignment Project

Lightweight Intrusion Detection System (IDS)

Protocols Monitored

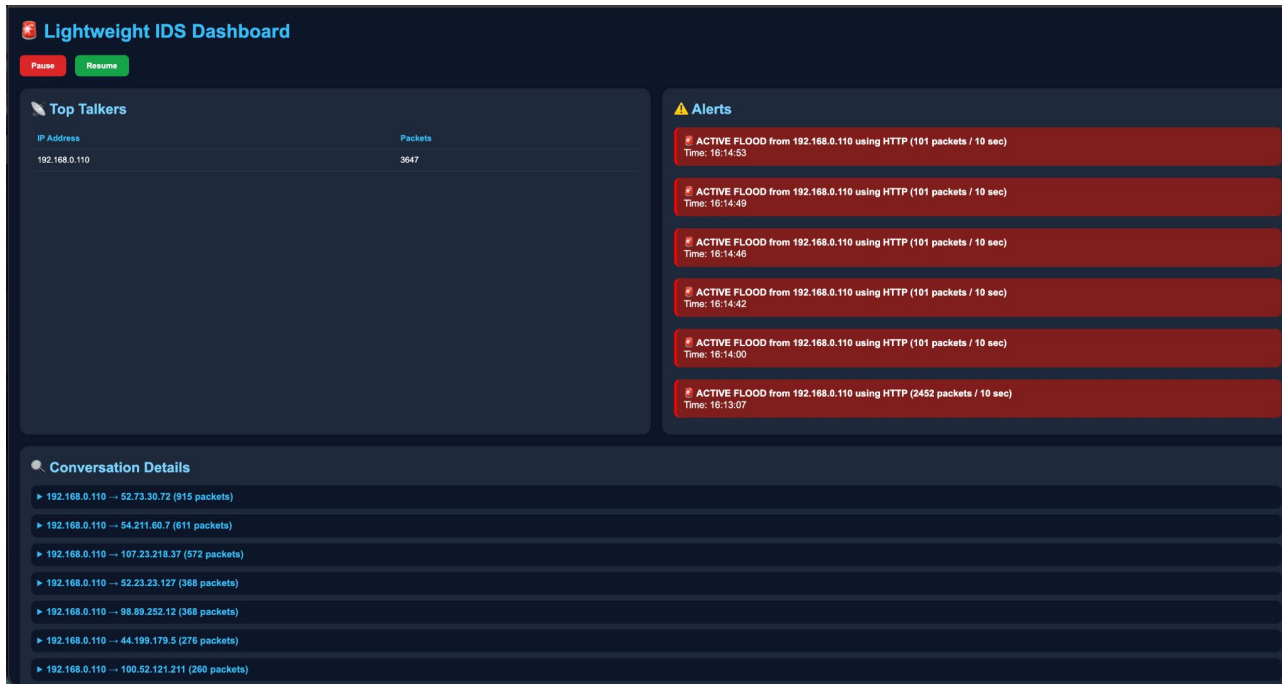
- HTTP (Port 80)
- FTP (Port 21)
- TELNET (Port 23)
- TFTP (Port 69)

Technologies Used

- Python
- Flask
- PyShark
- HTML/CSS/JavaScript

Key Features

- Live packet capture
- Top talker detection
- Real-time alerts
- Flood detection
- Protocol filtering
- Conversation tracking



Alert Suppression Logic

Cooldown timers prevent repeated false alerts from the same IP.

Benefits of the System

- Lightweight and fast
- Reduces false positives
- Focuses on insecure protocols
- Detects floods and traffic spikes in real time

Final Outcome

The IDS successfully:

- captured live traffic
- identified suspicious spikes
- detected flooding attacks
- tracked malicious source IPs
- visualized network behavior through a live dashboard

What is happening? (Flow)

- The IDS captures live network packets using PyShark.
- Only insecure protocols like HTTP, FTP, TELNET, and TFTP are monitored using port filtering.
- Source IP, destination IP, protocol, packet size, and payload preview are extracted from each packet.
- Packet timestamps are stored in a sliding 10-second window.
- The IDS calculates the current traffic rate for every source IP.
- Historical traffic data is used to learn the normal baseline behavior of each IP.
- If the current packet rate suddenly becomes much higher than the baseline, an anomaly alert is generated.
- If traffic exceeds a fixed threshold (e.g. 100 packets in 10 seconds), a flood alert is raised.
- Alert suppression timers are used to reduce repeated false alarms.
- All alerts, top talkers, conversations, and packet details are displayed live on the dashboard.

Some traffic generating checkers

Flood simulating links:

→ Continuous http flood:

```
while true
```

```
do
```

```
    curl -s http://httpbin.org/get > /dev/null
```

```
done
```

→ Heavy http flood(burst traffic)

```
seq 1 500 | xargs -n1 -P100 curl -s http://httpbin.org/get > /dev/null
```

FTP flood simulating link:

```
seq 1 300 | xargs -n1 -P50 curl -s ftp://ftp.gnu.org > /dev/null
```